SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of india)
Re-Accredited by NAAC with 'A++' Grade

Name: Aashray Lal Kayastha

PRN: 25070122005

Division: A1

ASSIGNMENT No: 6

TITLE: Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory

An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling

Outcome

Successfully implemented the use of inner class and an anonymous class

```java
class Outer {
    private String message = "Hello from Outer class";


    class Inner {
        void display() {
            System.out.println("Inner class accessing: " + message);
        }
    }
}

interface Greet {
    void sayHello();
}

public class JAVA_EXP_6 {
    public static void main(String[] args) {

        Outer outer = new Outer();
        Outer.Inner inner = outer.new Inner();
        inner.display();


        Greet g = new Greet() {
            public void sayHello() {
                System.out.println("Hello from Anonymous class");
            }
        };
        g.sayHello();
    }
}
```

```
C:\Users\Aashray\OneDrive\Desktop\JAVA_PRACTICALS>java Main.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Inner class accessing: Hello from Outer class
Hello from Anonymous class
```

Conclusion:

Therefore, from the experiment we got to learn how to use an inner class and an anonymous class.

Exercise

1.Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

Source code:

```java
class Vehicle {

    String name = "Car";
    String model = "Toyota";
    int year = 2024;

    // Inner Class
    class VehicleDetails {
        void display() {
            System.out.println("Vehicle Name : " + name);
            System.out.println("Model        : " + model);
            System.out.println("Year         : " + year);
        }
    }

    // Interface for Anonymous Class
    interface Action {
        void perform();
    }

    public static void main(String[] args) {

        Vehicle v = new Vehicle();

        // Inner class object
        VehicleDetails details = v.new VehicleDetails();
        details.display();

        // Anonymous class
        Action a = new Action() {
            public void perform() {
                System.out.println("Vehicle is running.");
            }
        };

        a.perform();
    }
}
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment6_Vehicle.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Vehicle Name : Car
Model        : Toyota
```

2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

Source code:

```
class FoodDelivery {

    String customer = "Rahul";
    String food = "Pizza";
    int quantity = 2;

    // Inner Class
    class OrderDetails {
        void displayOrder() {
            System.out.println("Customer Name : " + customer);
            System.out.println("Food Item     : " + food);
            System.out.println("Quantity      : " + quantity);
        }
    }

    // Interface for Anonymous Class
    interface DeliveryStatus {
        void status();
    }

    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        FoodDelivery fd = new FoodDelivery();

        // Inner class object
        OrderDetails order = fd.new OrderDetails();
        order.displayOrder();

        // Anonymous class
        DeliveryStatus ds = new DeliveryStatus() {
            public void status() {
                System.out.println(x: "Order is out for delivery.");
            }
        };

        ds.status();
    }
}
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment6_FoodDelivery.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Customer Name : Rahul
Food Item     : Pizza
Quantity      : 2
Order is out for delivery.
```